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Longitudinal characterization of the adipose tissue metabolome in dairy cows during the transition from cessation to resumption of lactation

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ABSTRACT

Dairy cows undergo profound metabolic challenges as they transition from lactation cessation to lactation resumption. Adipose tissue (AT), serving as the primary energy reserve and an active endocrine organ, plays a crucial role in these adaptations. Thus, the objective of the current study was a comprehensive characterization of the metabolic changes in the AT metabolome of Holstein dairy cows as they transitioned from one lactation cycle to the next, providing key insights into the dynamic adaptations crucial for maintaining energy homeostasis and optimizing lactational performance. Twelve Holstein dairy cows (BW = 745 ± 71 kg, BCS = 3.43 ± 0.66), housed in tiestalls, were dried off 6 wk before their expected calving date (mean dry-off time = 42 d). Cows were individually fed ad libitum TMR, consisting of grass silage, corn silage, and concentrate during lactation and a mixture of corn silage, barley straw, and concentrate during the dry period. The metabolome was characterized in subcutaneous AT samples collected on wk −7 (before drying off), −5 (after drying off), and wk −1 and 1 relative to calving. A targeted metabolomics approach was employed using the MxP Quant 500 kit (Biocrates Life Sciences AG), integrating liquid chromatography, flow injection, and electrospray ionization triple quadrupole MS. Statistical analysis of the AT metabolite data was conducted using MetaboAnalyst 5.0, enabling various multivariate analyses, including principal component analysis, partial least squares discriminant analysis, hierarchical clustering, and the generation of informative heatmaps. Multivariate analyses revealed distinct and dynamic alterations in the AT metabolome, with minimal changes during the early dry period (wk −7 to −5) followed by pronounced metabolic reprogramming close to calving (wk −1 to 1). Amino acid profiles in AT remained stable during late gestation but declined significantly in Ala, Asp, and Gln between wk −1 and wk 1, likely due to increased utilization within AT, redirecting carbon skeletons from these AA toward glyceroneogenesis and the re-esterification of fatty acids (FA) into triglycerides. Such a shift in AA metabolism may also facilitate interorgan nutrient exchange, with Ala export through the glucose-Ala cycle providing essential gluconeogenic substrates to the liver during early lactation. Moreover, acylcarnitine profiles remained unchanged, reflecting the role of AT as a long-term lipid reservoir rather than a metabolically active site for FA oxidation. The data revealed a biphasic pattern in diglycerides and extensive remodeling of phosphatidylcholines, underscoring dynamic cellular membrane adaptations to heightened lipolytic activity and increased energy demands during the immediate postpartum period. Notably, sphingomyelin remained stable throughout the transition, suggesting potential mechanisms in preserving membrane integrity and ensuring cellular stability under fluctuating metabolic stress. Together, these data further support that AT functions not merely as a passive energy store but as a dynamic organ actively orchestrating metabolic homeostasis during the transition from lactation cessation to lactation resumption.

Key words: adipose tissue, targeted metabolomics, transition, dairy cow

INTRODUCTION

Dairy science research has long focused on the transition period, the 3 wk before and after calving in dairy cows (Grummer, 1995). To support fetal growth, mammary tissue development, and the onset of lactation, this period is associated with significant changes in metabolic, endocrine, and immune functions (Grummer, 1995; Drackley, 1999; Kuhla et al., 2011). The abrupt increase in nutrient requirements of high-yielding dairy

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cows during late pregnancy and early lactation is largely supported by mobilization of body fat, protein, and glycogen (Drackley, 1999; Ingvartsen and Andersen, 2000; Sadri et al., 2023). This weight loss during the postpartum period appears to be an intrinsic trait, likely with a genetic basis (Friggens et al., 2007; Zachut et al., 2013; Zachut and Moallem, 2017). The extent of successful adaptation varies among individuals, with compromised adaptation predisposing cows to metabolic disorders (Drackley, 1999). Moreover, growing evidence suggests that the metabolic and immunological changes associated with drying off also affect the successful initiation and maintenance of lactation in dairy cows (Mezzetti et al., 2019). It has thus been proposed that the definition of the transition period be broadened to encompass the dry period (Mezzetti et al., 2020).

White adipose tissue (**AT**), the primary energy store in mammals, is essential for the successful establishment and maintenance of lactation in high-yielding dairy cows (Häussler et al., 2022). In addition to serving as an energy store, AT is now recognized as an active endocrine organ that secretes messenger molecules regulating inflammation, lipid homeostasis, insulin sensitivity, glucose metabolism, and fat distribution (Kershaw and Flier, 2004; Coelho et al., 2013; Häussler et al., 2022). Recent MS-based metabolomics studies have shown that AT in humans contains a complex network of metabolites, including branched-chain AA (**BCAA**), tricarboxylic acid (**TCA**) cycle intermediates, glycolytic compounds, and diverse lipids, which serve as potent markers of metabolic alterations (Newgard et al., 2009; Böhm et al., 2014; Kučera et al., 2018). For example, Koay et al. (2022) identified distinct omics signatures associated with insulin resistance and abdominal fat in obesity, whereas Mathioudaki et al. (2024) applied untargeted metabolomics to subcutaneous AT in type 2 diabetes, underscoring alterations in energy metabolism and novel biomarker profiles.

Research using metabolomics in dairy cow AT remains limited. To our knowledge, only one study has investigated lipidomics in dairy cow AT in relation to ketosis (Zhao et al., 2024), and another has investigated AT steroid metabolomics in periparturient dairy cows with varying BCS (Schuh et al., 2022). Most available AT metabolomics data, however, originate from studies on beef cattle. For example, Artegoitia et al. (2019) and Wang et al. (2024) reported breed-specific differences in lipid metabolism within perirenal AT, underscoring significant variability in fat deposition and metabolic efficiency. Additionally, Yamada et al. (2022) identified potential biomarkers linked to intramuscular adipogenesis through metabolomic analyses of intramuscular AT in Wagyu versus Holstein cattle. To our knowledge, no study has examined the AT metabolome in dairy cows during the transition from lactation cessation to lactation resumption; thus, the current study represents the first comprehensive report.

The objective of the present study was to comprehensively characterize the alterations in the AT metabolome in Holstein dairy cows before and during dry-off and around calving. To achieve this, we employed a targeted metabolomic approach, which enables precise quantification of known metabolites within key metabolic pathways. This approach was chosen over untargeted analysis to focus on specific metabolites with established physiological significance, allowing for a more refined investigation into AT metabolism. We hypothesized that this comprehensive characterization would reveal distinct metabolic signatures reflecting the adaptive responses necessary to overcome negative nutrient balance and support lactation. Such data will ultimately provide new insights into the interplay between lipid mobilization, endocrine function, and metabolic homeostasis during these critical periods.

MATERIALS AND METHODS

Animals, Experimental Design, and Sampling

The study was carried out between September 2019 and April 2020 at the Dairy Research Facility of Trouw Nutrition Research & Development (Boxmeer, the Netherlands). All procedures described in this manuscript fully complied with the Dutch Law on Experimental Animals, which aligns with ETS123 (Council of Europe 1985 and the 86/609/EEC Directive), and received approval from the Central Authority for Scientific Procedures on Animals (CCD, the Netherlands). The animals were part of an experiment designed to assess digestive and metabolic efficiency of energy and nitrogen utilization during both the lactation and dry periods in dairy cows. A complete description of the experimental design and performance data has been described previously (Daniel et al., 2022).

In brief, 12 Holstein Friesian cows were included in the study. Half of the cows were at the end of their first lactation, whereas the remaining cows were in their second to fifth lactations (3.2 ± 1.2; mean ± SD). At ~10 wk before their expected calving dates, the animals were moved from freestall to a tiestall equipped with individual feeding troughs and free access to water. During lactation, the cows were fed a TMR ad libitum, composed of 41.5% grass silage, 29.1% corn silage, and 29.4% mash compound feed. After a 2-wk acclimatization period, data collection and sampling activities were initiated. At dry-off, ~6 wk prepartum (median dry period lasted 43 d, with a range of 35 to 54 d), milking was stopped abruptly, and an internal teat sealant (OrbeSeal, Zoetis) was applied. The cows then received a dry cow TMR ad libitum, composed of 50.4% corn silage, 34.3% barley straw, and 15.3% mash compound feed. The chemical composition of the forages and concentrates and the calculated composition of the TMR have been previously reported (Daniel et al., 2022).

Subcutaneous AT samples were biopsied from the tail head region at wk −7, −5, −1, and 1 relative to calving. Before the biopsy, the cows were sedated by an intravenous injection of 0.5 mL xylazine (20 mg/mL; Sedazine, Produlab Pharma B.V.). The designated surgical area, previously shaved, was cleansed thoroughly by applying 3 alternating rounds of a povidone-iodine scrub, each followed by a rinse with 70% isopropyl alcohol. After disinfection of the surgical site, local anesthesia was then achieved by administering 6 mL of procaine hydrochloride (20 mg/mL; Procamidor, Richter Pharma AG, Wels, Austria) anterior to the biopsy site. A 2-cm incision was made through the skin using a scalpel and ~1 g of subcutaneous AT was collected. The incision site was promptly closed using 6 to 8 skin staples (Autosuture Royal 35W skin stapler, Covidien, Ireland), and an aerosol containing 4% aluminum powder (Aluspray, Vetoquinol, the Netherlands) was applied to form a protective barrier against contaminants. Staples were removed after a minimum of 10 d. The tissue sample was immediately transferred to a Petri dish, rinsed with sterile saline, and then split into 3 pieces. Each piece was placed into a separate cryovial, snap-frozen in liquid nitrogen, and stored at −80°C until analysis. After the procedure, cows were given an intramuscular injection of ketoprofen (100 mg/mL; Ceva Santé Animale B.V.) at a dosage of 3 mL per 100 kg BW.

Metabolomics Analysis and Data Processing

For targeted metabolomics analyses, AT samples were processed using the extraction procedure provided by Biocrates Life Sciences AG (Innsbruck, Austria), which is based on the methodology detailed by Zukunft et al. (2018). In brief, 100 mg of AT were homogenized in ethanol/phosphate buffer mixture (85:15, with a tissue-to-solvent ratio of 1:6 wt/vol) in a 2 mL Precellys CK14 tube containing 1.4 mm ceramic (zirconium oxide) beads using a Precellys 24 tissue homogenizer (Bertin Technologies SAS, Montigny-le-Bretonneux, France) at 4°C. Three consecutive homogenization cycles of 20-s each were performed at 5,500 rpm, with 30-s pauses between cycles. Following homogenization, the samples were cooled on ice for 20 min with inversion every 5 min to maintain homogeneity. The homogenates were then centrifuged at 10,000 × *g* for 5 min at 4°C to obtain a uniform supernatant for subsequent analyses.

The metabolite concentrations were determined using the MxP Quant 500 kit, following the manufacturer's instructions (Biocrates Life Sciences AG). In brief, 10 μL of the supernatant were transferred to the well plate of the kit. Subsequent analyses were performed via tandem MS at the Fraunhofer Institute for Toxicology and Experimental Medicine (Hannover, Germany), following the detailed method described by Mahajan et al. (2020). For lipid quantification, flow injection analysis-tandem MS (**FIA-MS/MS**) was used on a 5500 QTRAP mass spectrometer (AB Sciex, Darmstadt, Germany) equipped with an electrospray ionization source. Small molecule metabolites were assayed by liquid chromatography-MS (**LC-MS/MS**) using an Agilent 1290 Infinity II system (Santa Clara, CA) coupled with a tandem mass spectrometer. Detection of analytes in both workflows was achieved through multiple reaction monitoring. Sample preparation was carried out using 96-well plate devices that contained internal standards and a predetermined number of samples. In the case of some analytes, such as AA, derivatization was performed by adding a phenyl isothiocyanate solution. Thereafter, the target metabolites were extracted using an organic solvent and subsequently diluted.

Metabolite data were initially quantified using Sciex Analyst MS software (version 1.7.2 Build 8287) and then imported into Biocrates MetIDQ software (Oxygen-DB110–3023) for analyte identification, concentration calculations, and data compilation. For the LC-MS/MS measurements, quantification was performed by stable isotope dilution combined with 7-point calibration curves. In the case of the FIA-MS/MS assay, metabolite concentrations were calculated using a one-point calibration based on the internal standard, which was also isotopically corrected.

In compliance with the manufacturer's guidelines, targeted metabolomic data were processed and managed using MetIDQ Boron software. To determine the limits of detection (**LOD**), blank PBS samples were analyzed in triplicate. The background noise for each metabolite was determined by calculating the median values of all PBS samples on the plate. Following the Biocrates standard operating procedure, the LOD were set at 3 times the signal obtained from the blank sample that was extracted and analyzed immediately before the sample runs. Any metabolite values falling below this threshold were considered as missing. Metabolite concentrations in AT were expressed in μmol/mg of tissue and were normalized using internal quality control samples. The MxP Quant 500 kit allowed the quantitative analysis of metabolites from 12 biochemical classes of lipids and an additional 14 classes of small molecules.

Statistical Analyses

MetaboAnalyst 6.0 (http://www.metaboanalyst.ca), a web-based tool for processing metabolomic data, was employed for the statistical evaluation of the AT metabolite data. Before analysis, we carefully checked the .csv file to make sure that it contained only numeric values and followed standard formatting guidelines. As part of the data filtering process, we applied an interquartile range (**IQR**) assessment to identify and remove outliers. The IQR, calculated as the difference between the 75th and 25th percentiles, is a statistical measure that indicates the variation in a dataset. Any observation falling outside the interval defined by (25th percentile − 1.5 × IQR) and (75th percentile + 1.5 × IQR) was noted as an outlier (Dash et al., 2023). To ensure robustness and reliable statistical analysis, we discarded variables exhibiting more than 5% missing values (Pang et al., 2024). For the remaining missing values, the k-nearest neighbors algorithm was used to impute data, thereby maintaining the integrity of the dataset. Subsequently, a generalized logarithm transformation (base 10) was applied, followed by Pareto scaling to correct for heteroscedasticity, reduce skewness, and minimize masking effects. The processed data were then subjected to various multivariate analyses, including principal component analysis (**PCA**), partial least squares discriminant analysis (**PLS-DA**), hierarchical clustering, and the generation of informative heatmaps (Pang et al., 2024). For PCA, differences in metabolic profiles across time points were evaluated by permutational multivariate ANOVA (**PERMANOVA**; 999 permutations) on the PCA scores. The *F*-value, R^2, and *P*-value were used to quantify statistical significance and effect size of temporal variation. The PLS-DA, a supervised classification method incorporating class labels in the analysis, was used to identify components that maximize the separation between predefined groups in the data. The PLS-DA model was validated using cross-validation and permutation testing, with R^2 and Q^2 values serving as indicators of the model's explanatory and predictive power, respectively. Additionally, univariate analyses were carried out through fold-change analysis and volcano plots to identify significantly affected metabolites during 3 critical periods: from wk −7 to −5 (during the transition from lactation to nonlactation), from wk −5 to wk −1 (during the dry period), and from wk −1 to wk 1 (during the transition from dry period to the next lactation). For volcano plots, a fold-change threshold of 1.5 along the x-axis and a *P*-value threshold of 0.05 along the y-axis were applied. Before conducting these analyses, Q-Q plots and residual plots were examined to

Volcano plots were also employed to compare AT metabolites between primiparous and multiparous cows, focusing on sampling times (wk −7, −5, −1, and 1 relative to calving). The analysis revealed no significant differences in metabolite profiles between the 2 groups across all time points. Consequently, parity was not considered a factor in the subsequent analysis of AT metabolites.

RESULTS AND DISCUSSION

The PCA analysis revealed that the first 3 components, accounting for ~ 35.5%, 17.1%, and 8.3% of the variance (Figure 1A), captured dynamic changes in the AT metabolome during the transition from lactation cessation to lactation resumption. The observed group separation was statistically supported by PERMANOVA analysis (F = 4.50, R^2 = 0.24, P = 0.001; 999 permutations). The clustering of samples based on the time of sampling (wk), particularly the clear separation of wk −7 and wk 1 from wk −5 and wk −1, indicates pronounced metabolic reprogramming in response to the changing physiological demands around calving. The PLS-DA further enhanced the separation among sampling time points (Figure 1B). The robustness of this model is underscored by high cross-validation metrics (R^2 = 0.92 and Q^2 = 0.78 for 4 components; Figure 1C) and a significant permutation test result ($P = 5 \times 10^{-4}$; Figure 1D), confirming that the metabolic changes observed are not merely random but are strongly associated with the transition from lactation cessation to lactation resumption.

It should be noted that volcano plots in Figures 2, 3, 4, 5, 6, and 7 are not presented for comparisons where no significantly altered metabolites were identified. Supplemental Figure S1 (see Notes) provides a detailed visualization of these comparisons, demonstrating that AT metabolites showed no significant changes across the following contrasts relative to calving: wk −5 to −7 (AA, acylcarnitines, phosphatidylcholines [**PC**], and sphingomyelines [**SM**]), wk −5 to −1 (AA, biogenic amines, acylcarnitines, and SM), and wk −1 to 1 (acylcarnitines and SM).

AA

The heatmap in Figure 2A shows temporal changes in AA concentrations in AT across sampling points. Prepartum samples (wk −7, −5, and −1) displayed fairly stable AA profiles, whereas a marked shift was observed by wk 1 postpartum, suggesting that calving triggers substantial reprogramming of AA metabolism to support lactation. Figure 2B displays a volcano plot comparing wk −1 to wk 1, revealing significant decreases in Ala, Asp, and Gln. In addition to storing lipids, AT functions as an endocrine gland that regulates glucose and lipid homeostasis through adipokine secretion (Häussler et al., 2022). However, its role in systemic protein and AA metabolism remains relatively understudied in ruminants. Several in

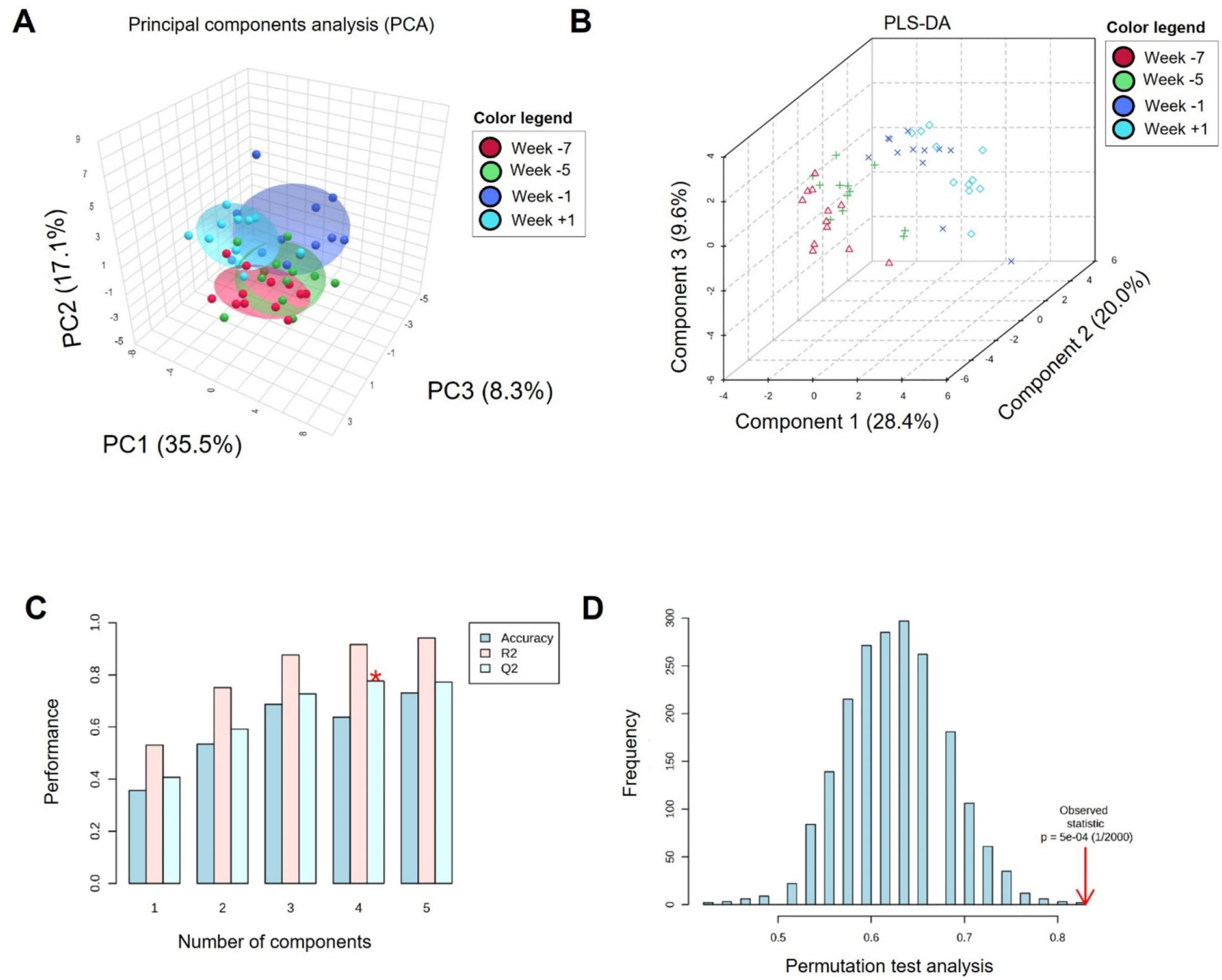

Figure 1. (A) Principal component analysis (PCA) score plot illustrating the relationship between principal components (PC) 1, 2, and 3 in the adipose tissue metabolome of dairy cows (n = 12). Each symbol represents the PC values of an individual cow at a specific sampling week. Symbols are color-coded to represent the different sampling weeks relative to parturition (wk −7, −5, −1, +1). (B) Two-dimensional score plots of partial least square discriminant analysis (PLS-DA), demonstrating the separation of the adipose tissue metabolome of dairy cows. The PLS-DA is a statistical approach that was used to classify and differentiate sampling time points based on adipose tissue metabolomic data. (C) Cross-validation results for PLS-DA, employing a leave-one-out cross-validation technique to assess model performance and generalizability. The R^2 and Q^2 values were checked to assess the model's explanatory and predictive power, respectively. (D) Permutation test results, assessing prediction accuracy during training. The permutation test is a statistical method used to validate the significance of the model's predictive performance by randomly shuffling data labels and assessing the resulting accuracy.

vitro and in vivo studies suggest that AT is metabolically active in the context of AA metabolism (Frayn et al., 1991; Kowalski and Watford, 1994; Kowalski et al., 1997). In AT, BCAA are considered the primary nitrogen source for synthesizing Ala and Gln (Frayn et al., 1991; Kowalski et al., 1997). In fact, Glu produced from the deamination of BCAA can be used to synthesize nonessential AA such as Ala and Gln (Tischler and Goldberg, 1980) and, via transamination with oxaloacetate, Asp, with BCAA catabolism enhancing TCA cycle flux. Supporting this, studies in mice (Herman et al., 2010), sheep (Bergen et al., 1988), and dairy cows (Liang et al., 2019; Webb et al., 2019, 2020) indicate that AT plays an important role in BCAA metabolism, with subcutaneous AT serving as a key site for BCAA uptake and initial catabolism in dairy cows (Webb et al., 2019, 2020). Assuming that these 3 AA (Ala, Gln, and Asp) originate from BCAA catabolism in AT, their reductions could result from multiple factors, including insufficient dietary BCAA intake, diminished transport into AT (potentially to spare BCAA for milk

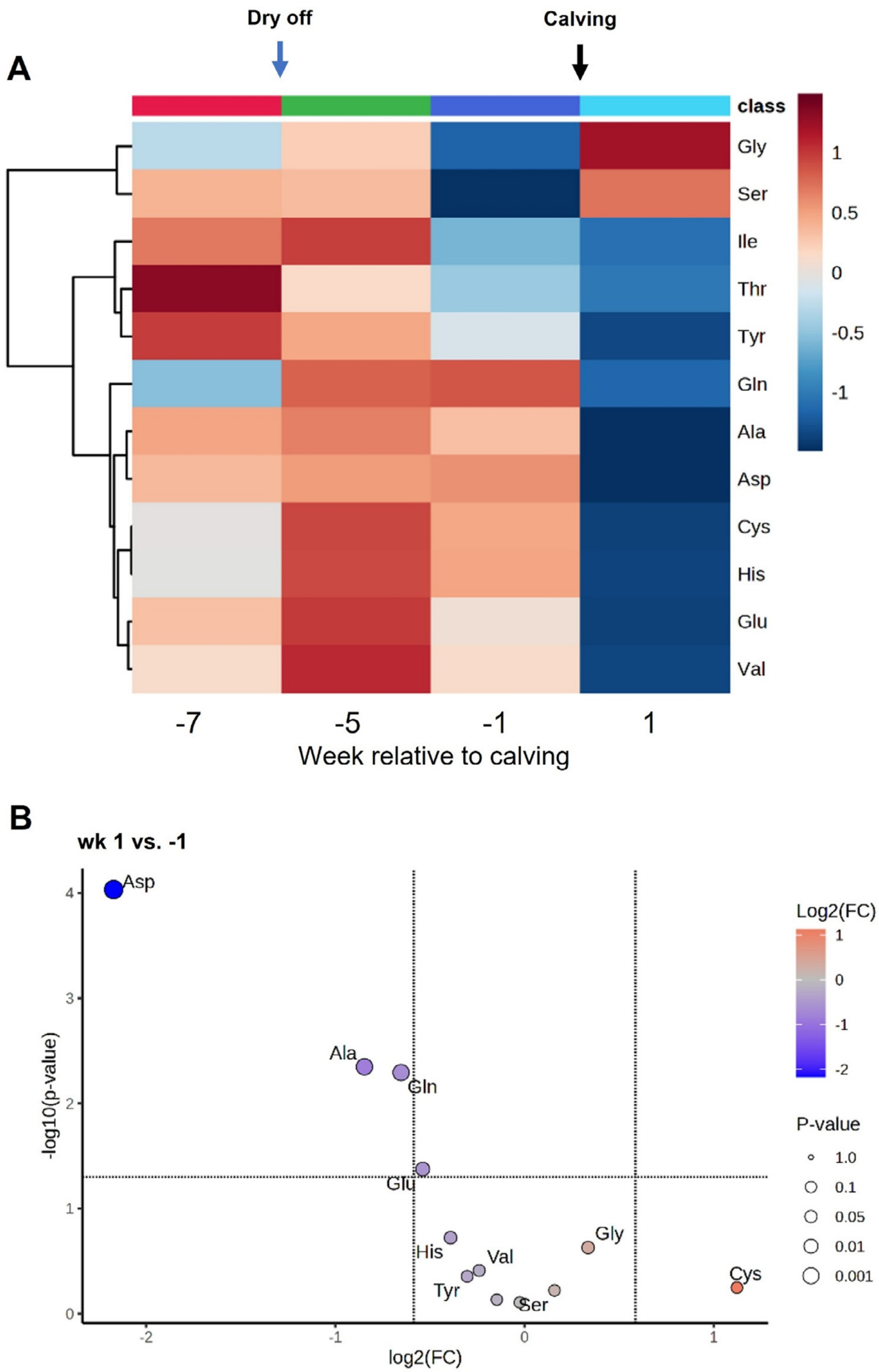

Figure 2. (A) Heatmap illustrating the changes in adipose tissue AA over the sampling period. Subcutaneous adipose tissue samples were collected from 12 Holstein Friesian cows on wk −7, −5, −1, and 1 relative to calving. The targeted metabolomics approach was conducted using the MxP Quant 500 kit (Biocrates Life Sciences AG), integrating liquid chromatography, flow injection, and electrospray ionization triple quadrupole MS. The color gradient represents mean centered metabolite concentrations, scaled within the range of each variable. (B) Volcano plots visualizing adipose tissue metabolites that differed between week (wk) 1 and −1 relative to calving. The x-axis represents the mean $\log_2$ fold-change (FC), while the y-axis displays the negative logarithm of the P-values. Each circle on the plot corresponds to an individual metabolite. The t-tests were performed with a threshold of $P < 0.05$ to determine the level of significance, represented by the dashed horizontal line. The threshold set for fold change was 1.5, which is indicated by the vertical lines. The volcano plot did not reveal any significantly altered metabolites between wk −7 and −5, as well as between wk −1 and −5, and therefore these comparisons are not displayed.

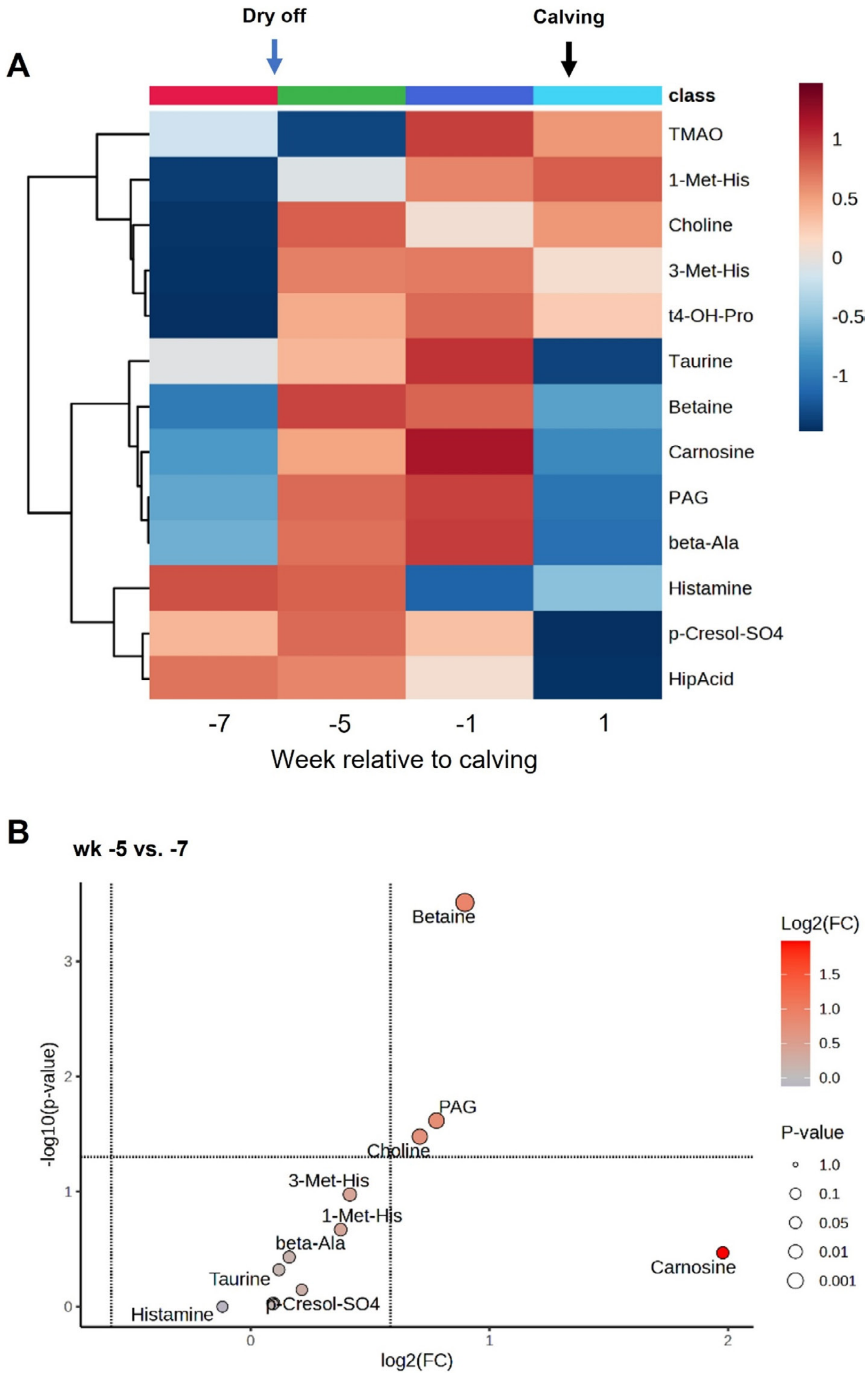

Figure 3. (A) Heatmap illustrating the changes in adipose tissue biogenic amines and AA-related metabolites over the sampling period. Subcutaneous adipose tissue samples were collected from 12 Holstein Friesian cows on wk −7, −5, −1, and 1 relative to calving. The targeted metabolomics approach was conducted using the MxP Quant 500 kit (Biocrates Life Sciences AG), integrating liquid chromatography, flow injection, and electrospray ionization triple quadrupole MS. The color gradient represents mean centered metabolite concentrations, scaled within the range of each variable. Volcano plots visualizing adipose tissue metabolites that differed between week (wk) −5 and −7 (B) and wk 1 and −1 (C) relative to calving. The x-axis represents the mean $\log_2$ fold-change (FC), while the y-axis displays the negative logarithm of the *P*-values. Each circle on the plot corresponds to an individual metabolite. The *t*-tests were performed with a threshold of $P < 0.05$ to determine the level of significance, represented by the dashed horizontal line. The threshold set for fold change was 1.5, which is indicated by the vertical lines. The volcano plot did not reveal any significantly altered metabolites between wk −5 and −1, and therefore this comparison is not displayed. PAG = phenylacetylglycine, p-cresol-SO4 = p-cresol sulfate, 3-Met-His = 3-methylhistidine, HipAcid = hippuric acid, t4-OH-Pro = trans-4-hydroxyproline.

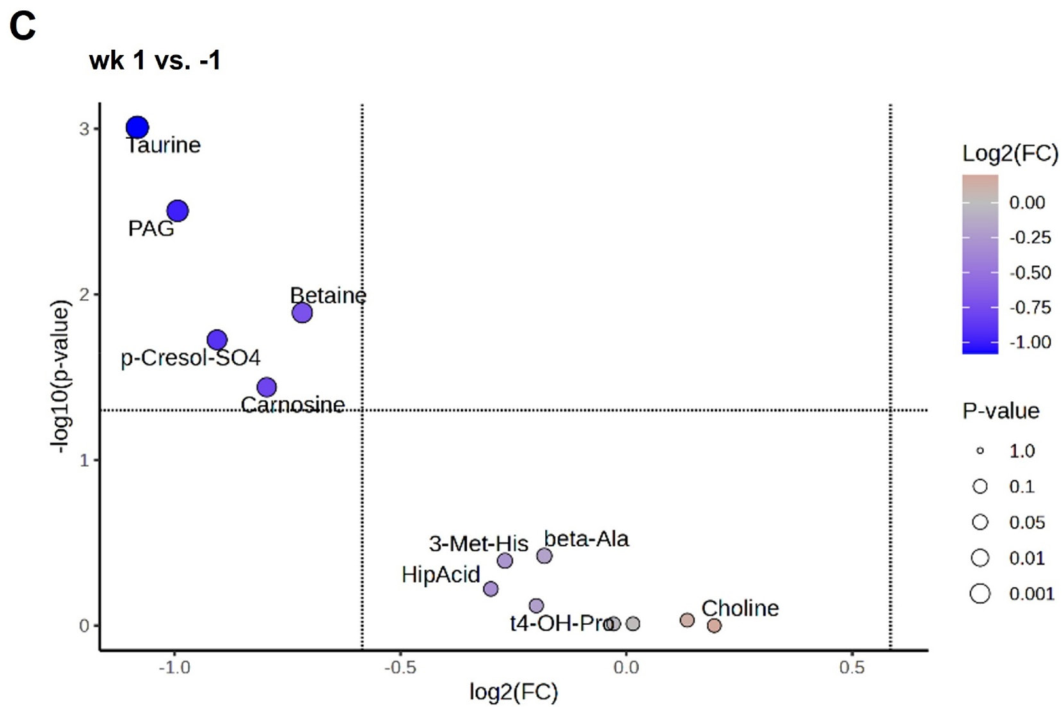

Figure 3 (Continued). (A) Heatmap illustrating the changes in adipose tissue biogenic amines and AA-related metabolites over the sampling period. Subcutaneous adipose tissue samples were collected from 12 Holstein Friesian cows on wk −7, −5, −1, and 1 relative to calving. The targeted metabolomics approach was conducted using the MxP Quant 500 kit (Biocrates Life Sciences AG), integrating liquid chromatography, flow injection, and electrospray ionization triple quadrupole MS. The color gradient represents mean centered metabolite concentrations, scaled within the range of each variable. Volcano plots visualizing adipose tissue metabolites that differed between week (wk) −5 and −7 (B) and wk 1 and −1 (C) relative to calving. The x-axis represents the mean $\log_2$ fold-change (FC), while the y-axis displays the negative logarithm of the *P*-values. Each circle on the plot corresponds to an individual metabolite. The *t*-tests were performed with a threshold of $P < 0.05$ to determine the level of significance, represented by the dashed horizontal line. The threshold set for fold change was 1.5, which is indicated by the vertical lines. The volcano plot did not reveal any significantly altered metabolites between wk −5 and −1, and therefore this comparison is not displayed. PAG = phenylacetylglycine, p-cresol-SO4 = p-cresol sulfate, 3-Met-His = 3-methylhistidine, HipAcid = hippuric acid, t4-OH-Pro = trans-4-hydroxyproline.

component synthesis), increased utilization within AT to meet physiological demands, or enhanced release from AT.

Of the 4 possibilities, the third, increased utilization within AT to meet physiological demands, suggests that the carbon skeletons of Ala, Gln, and Asp might serve as substrates for the production of 3-phosphoglycerol (**G3P**; Tordjman et al., 2007; Nye et al., 2008), which is essential for glyceroneogenesis and the re-esterification of fatty acids (**FA**) into triglycerides (Hanson and Reshef, 2003). This pathway provides a potential link between AA and FA metabolism, particularly in the context of glyceroneogenesis (Kainulainen et al., 2013). A large portion of hydrolyzed FA in adipocytes is re-esterified to triacylglycerol (**TG**) in a "futile," ATP-consuming, energy-dissipating cycle (Nye et al., 2008). Re-esterification protects the adipocytes from endoplasmic reticulum stress caused by free FA from lipolysis (Chitraju et al., 2017). Due to the lack of glycerol kinase, glycerol from lipolysis cannot be phosphorylated to G3P (Chilliard, 1993) but can be released through aquaporin-7 as the main gateway (Rodriguez et al., 2011) and used for gluconeogenesis in other organs such as the liver and the kidneys, albeit at low rates when compared with other substrates (Aschenbach et al., 2019). The decrease in aquaporin-7 mRNA with the onset of lactation in AT of dairy cows (Sauerwein et al., 2013) may indicate reduced release of glycerol.

Regarding the potential enhanced release of Ala, Gln, and Asp from AT, in vivo studies in humans and laboratory animals using both microdialysis (Kowalski and Watford, 1994) and arterio-venous difference measurements (Frayn et al., 1991; Kowalski et al., 1997) have shown that AT releases Ala and Gln while taking up Glu at rates that may give it an appreciable role in whole-body metabolism (Kim et al., 2008; Hanzu et al., 2014). During early lactation in cows, when other tissues such as the liver are prioritized in the supply of gluconeogenic substrates, the observed decrease in Ala may result from its direct export via the glucose-Ala cycle to support hepatic gluconeogenesis (Holeček, 2024; Kong et al., 2025). The glucose-Ala cycle, also

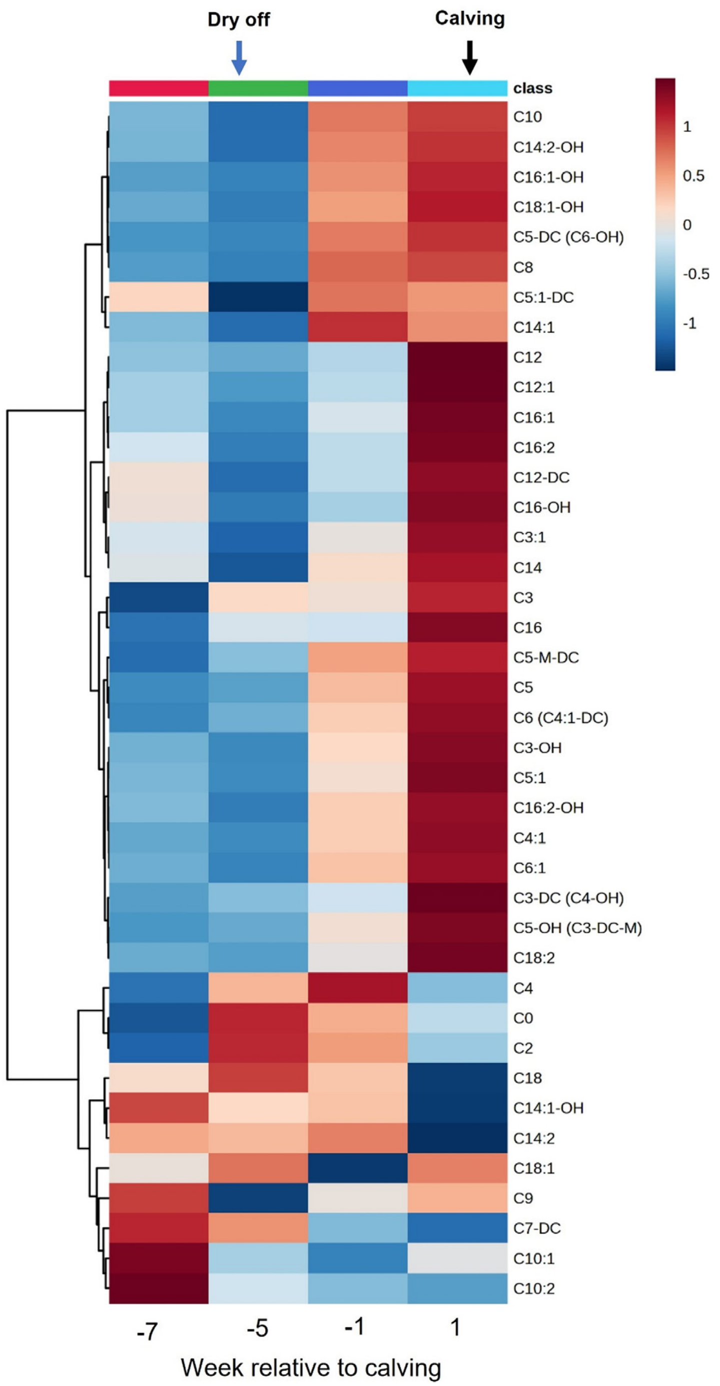

Figure 4. Heatmap illustrating the changes in adipose tissue acylcarnitines over the sampling period. Subcutaneous adipose tissue samples were collected from 12 Holstein Friesian cows on wk −7, −5, −1, and 1 relative to calving. The targeted metabolomics approach was conducted using the MxP Quant 500 kit (Biocrates Life Sciences AG), integrating liquid chromatography, flow injection, and electrospray ionization triple quadrupole MS. The color gradient represents mean centered metabolite concentrations, scaled within the range of each variable. C0 = carnitine, C2 = acetylcarnitine, C3 = propionylcarnitine, C3-DC = malonylcarnitine, C4-OH = hydroxybutyrylcarnitine, C3-OH = hydroxypropionylcarnitine, C3:1 = propenoylcarnitine, C4:0 = butyrylcarnitine, C5:0 = valerylcarnitine, C5-DC = glutarylcarnitine, C5-M-DC = Methylglutarylcarnitine, C5-OH = hydroxyvalerylcarnitine, C5:1 = tiglylcarnitine, C5:1-DC = glutaconylcarnitine, C6 (C4:1-DC) = hexanoylcarnitine, C6:1 = hexenoylcarnitine, C7-DC = pimeloylcarnitine, C8:0 = octanoylcarnitine, C9:0 = nonaylcarnitine, C10:0 = decanoylcarnitine, C10:1 = decenoylcarnitine, C10:2 = decadienoylcarnitine, C12:0 = dodecanoylcarnitine, C12-DC = dodecanedioylcarnitine, C12:1 = dodecenoylcarnitine, C14:0 = tetradecanoylcarnitine, C14:1 = tetradecenoylcarnitine, C14:1-OH = hydroxytetradecenoylcarnitine, C14:2 = tetradecadienoylcarnitine, C14:2-OH = hydroxytetradecadienoylcarnitine, C16:0 = hexadecanoylcarnitine, C16-OH = hydroxyhexadecanoylcarnitine, C16:1 = hexadecenoylcarnitine, C16:2 = hexadecadienoylcarnitine, C16:2-OH = hydroxyhexadecadienoyl-carnitine, C18:0 = octadecanoylcarnitine, C18:1 = octadecenoylcarnitine, C18:1-OH = hydroxyoctadecenoylcarnitine, C18:2 = octadecadienylcarnitine.

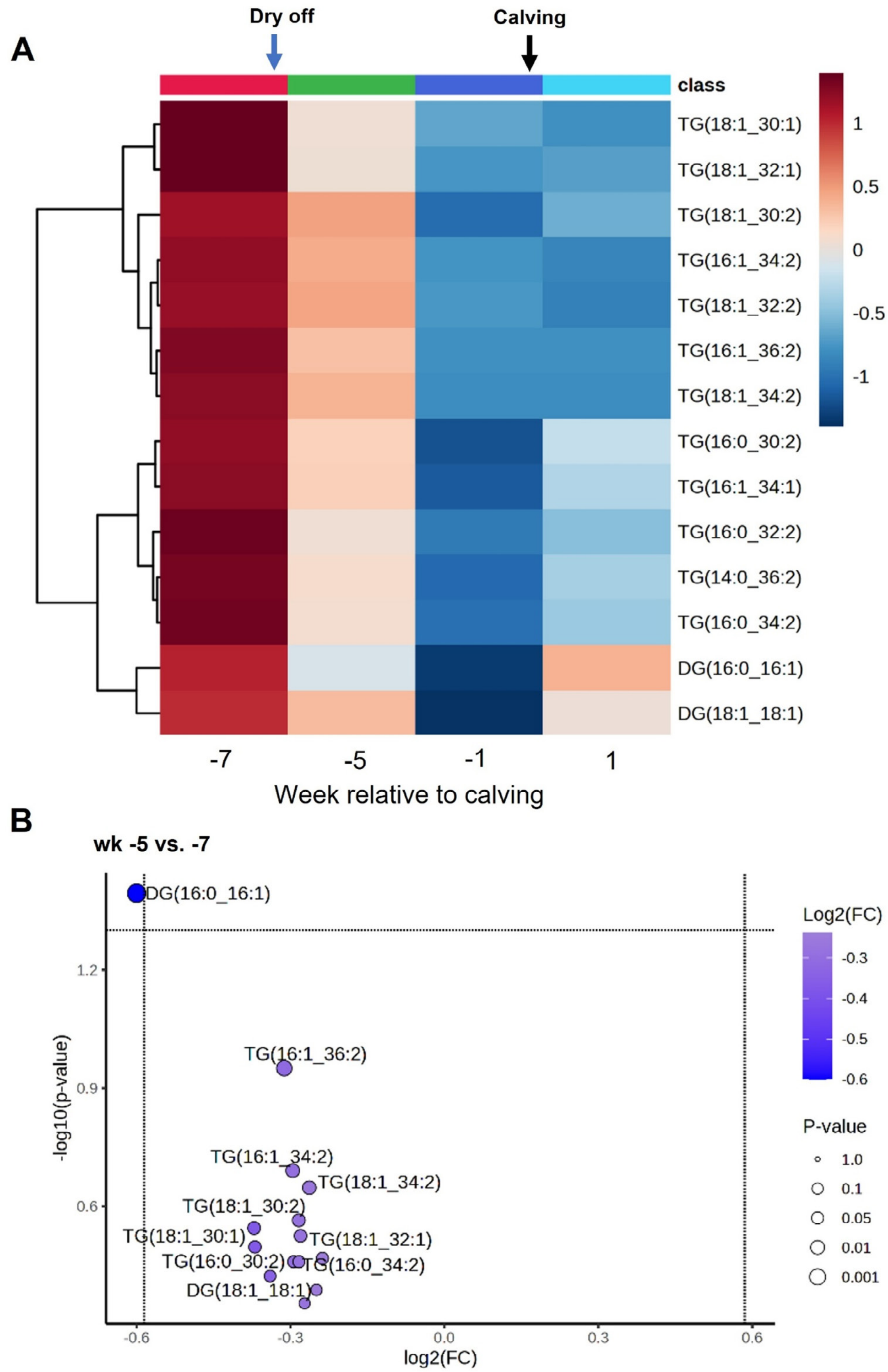

Figure 5. (A) Heatmap illustrating the changes in adipose tissue diglycerides and triglycerides over the sampling period. Subcutaneous adipose tissue samples were collected from 12 Holstein Friesian cows on wk −7, −5, −1, and 1 relative to calving. The targeted metabolomics approach was conducted using the MxP Quant 500 kit (Biocrates Life Sciences AG), integrating liquid chromatography, flow injection, and electrospray ionization triple quadrupole MS. The color gradient represents mean centered metabolite concentrations, scaled within the range of each variable. Volcano plots visualizing adipose tissue metabolites that differed between week (wk) −5 and −7 (B), wk −1 and −5 (C), and wk 1 and −1 (D) relative to calving. The x-axis represents the mean $\log_2$ fold-change (FC), while the y-axis displays the negative logarithm of the *P*-values. Each circle on the plot corresponds to an individual metabolite. The *t*-tests were performed with a threshold of $P < 0.05$ to determine the level of significance, represented by the dashed horizontal line. The threshold set for fold change was 1.5, which is indicated by the vertical lines.

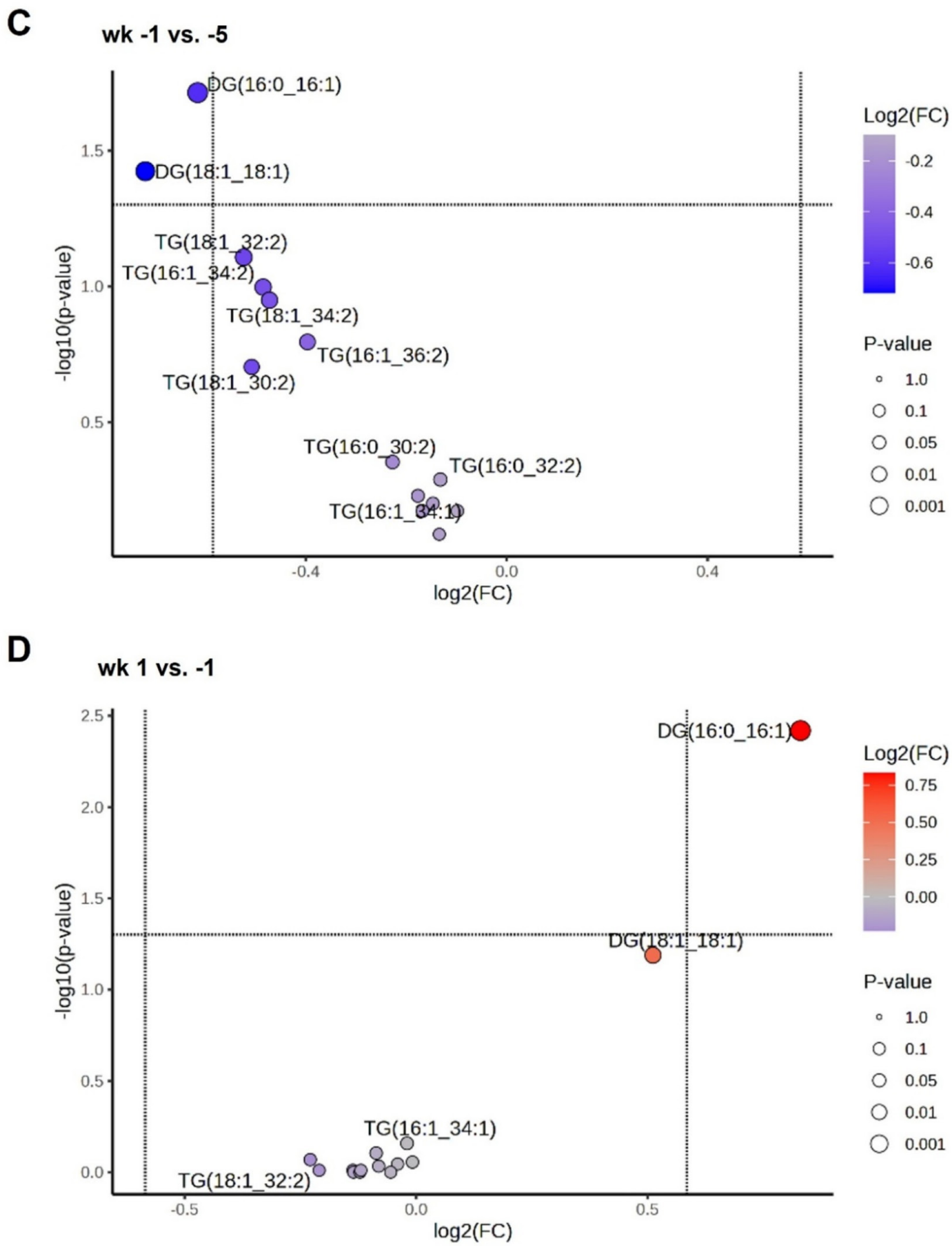

Figure 5 (Continued). (A) Heatmap illustrating the changes in adipose tissue diglycerides and triglycerides over the sampling period. Subcutaneous adipose tissue samples were collected from 12 Holstein Friesian cows on wk −7, −5, −1, and 1 relative to calving. The targeted metabolomics approach was conducted using the MxP Quant 500 kit (Biocrates Life Sciences AG), integrating liquid chromatography, flow injection, and electrospray ionization triple quadrupole MS. The color gradient represents mean centered metabolite concentrations, scaled within the range of each variable. Volcano plots visualizing adipose tissue metabolites that differed between week (wk) −5 and −7 (B), wk −1 and −5 (C), and wk 1 and −1 (D) relative to calving. The x-axis represents the mean $\log_2$ fold-change (FC), while the y-axis displays the negative logarithm of the *P*-values. Each circle on the plot corresponds to an individual metabolite. The *t*-tests were performed with a threshold of $P < 0.05$ to determine the level of significance, represented by the dashed horizontal line. The threshold set for fold change was 1.5, which is indicated by the vertical lines.

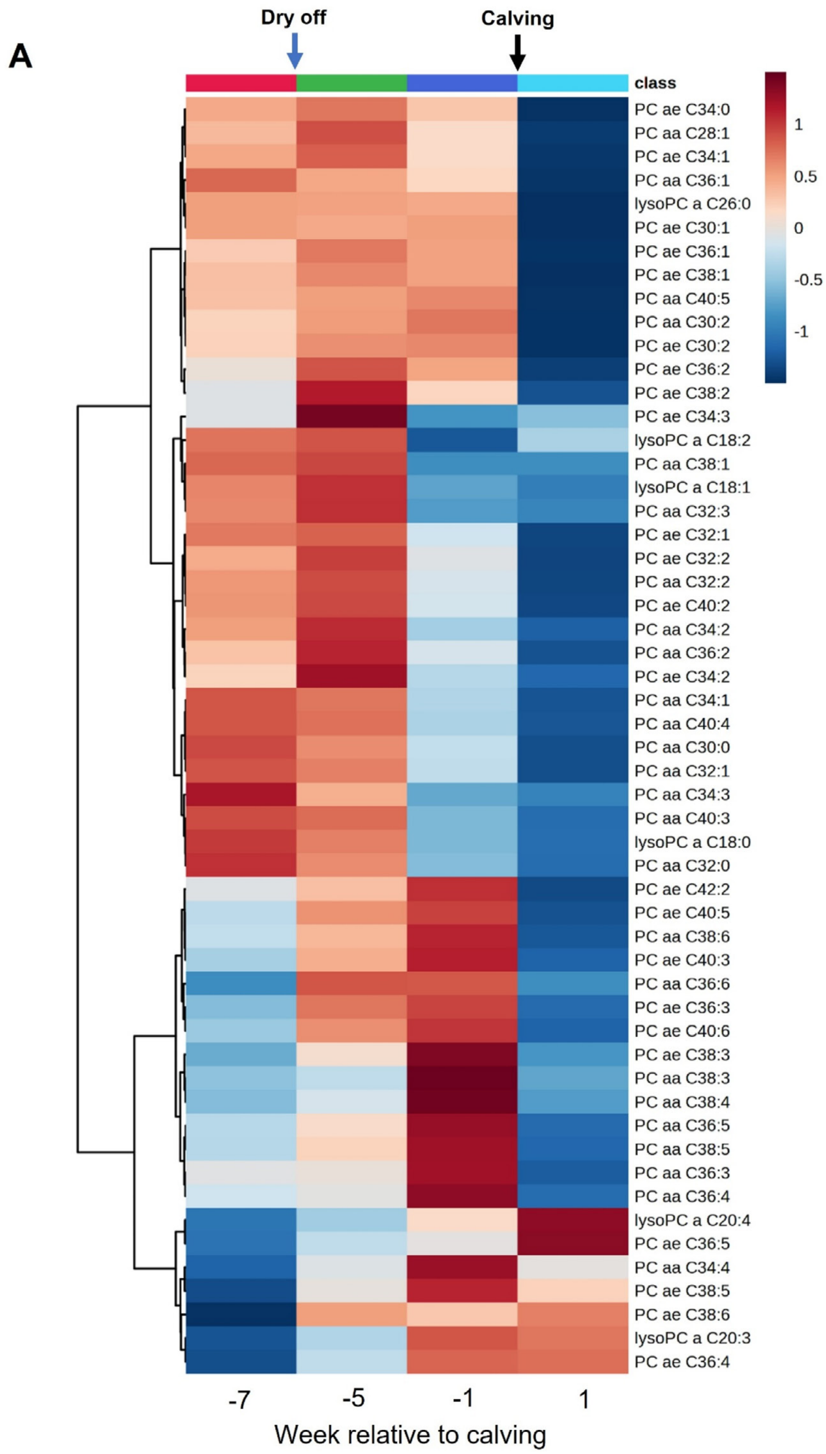

Figure 6. (A) Heatmap illustrating the changes in adipose tissue acyl alkyl phosphatidylcholines (PC ae), di-acyl phosphatidylcholines (PC aa), and lysophosphatidylcholines (LysoPC) over the sampling period. Subcutaneous adipose tissue samples were collected from 12 Holstein Friesian cows on wk −7, −5, −1, and 1 relative to calving. The targeted metabolomics approach was conducted using the MxP Quant 500 kit (Biocrates Life Sciences AG), integrating liquid chromatography, flow injection, and electrospray ionization triple quadrupole MS. The color gradient represents mean centered metabolite concentrations, scaled within the range of each variable. Volcano plots visualizing adipose tissue metabolites that differed between week (wk) −1 and −5 (B), and wk 1 and −1 (C) relative to calving. The x-axis represents the mean $\log_2$ fold-change (FC), while the y-axis displays the negative logarithm of the *P*-values. Each circle on the plot corresponds to an individual metabolite. The *t*-tests were performed with a threshold of $P < 0.05$ to determine the level of significance, represented by the dashed horizontal line. The threshold set for fold change was 1.5, which is indicated by the vertical lines. The volcano plot did not reveal any significantly altered metabolites between wk −5 and −7, and therefore this comparison is not displayed.

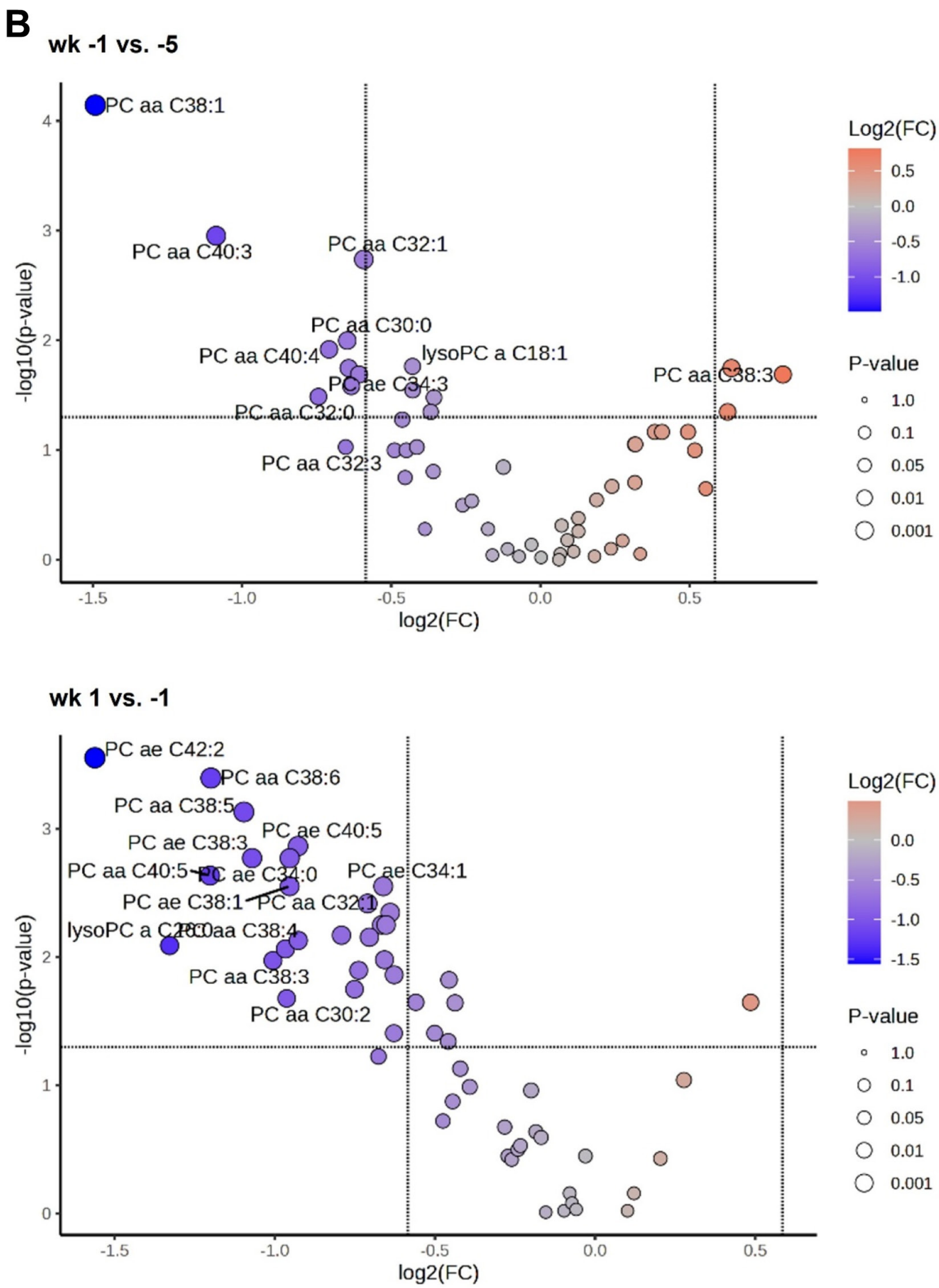

Figure 6 (Continued). (A) Heatmap illustrating the changes in adipose tissue acyl alkyl phosphatidylcholines (PC ae), di-acyl phosphatidylcholines (PC aa), and lysophosphatidylcholines (LysoPC) over the sampling period. Subcutaneous adipose tissue samples were collected from 12 Holstein Friesian cows on wk −7, −5, −1, and 1 relative to calving. The targeted metabolomics approach was conducted using the MxP Quant 500 kit (Biocrates Life Sciences AG), integrating liquid chromatography, flow injection, and electrospray ionization triple quadrupole MS. The color gradient represents mean centered metabolite concentrations, scaled within the range of each variable. Volcano plots visualizing adipose tissue metabolites that differed between week (wk) −1 and −5 (B), and wk 1 and −1 (C) relative to calving. The x-axis represents the mean $\log_2$ fold-change (FC), while the y-axis displays the negative logarithm of the *P*-values. Each circle on the plot corresponds to an individual metabolite. The *t*-tests were performed with a threshold of $P < 0.05$ to determine the level of significance, represented by the dashed horizontal line. The threshold set for fold change was 1.5, which is indicated by the vertical lines. The volcano plot did not reveal any significantly altered metabolites between wk −5 and −7, and therefore this comparison is not displayed.

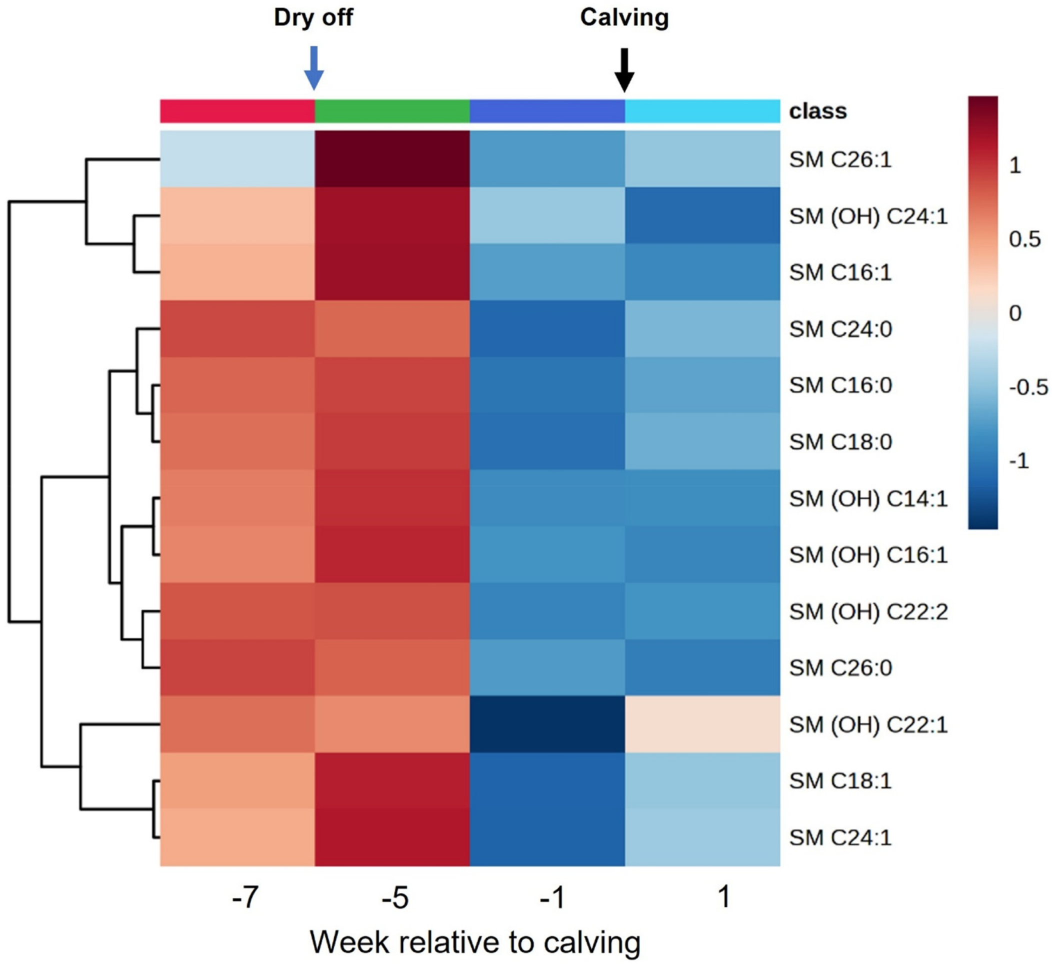

Figure 7. (A) Heatmap illustrating the changes in adipose sphingomyelins (SM) over the sampling period. Subcutaneous adipose tissue samples were collected from 12 Holstein Friesian cows on wk −7, −5, −1, and 1 relative to calving. The targeted metabolomics approach was conducted using the MxP Quant 500 kit (Biocrates Life Sciences AG), integrating liquid chromatography, flow injection, and electrospray ionization triple quadrupole MS. The color gradient represents mean centered metabolite concentrations, scaled within the range of each variable. The volcano plots did not reveal any significantly altered metabolites between wk −7 and −5, wk −5 and −1, and wk −1 and 1, which were not depicted here.

known as the Cahill cycle, is a vital metabolic pathway linking carbohydrate and AA metabolism. In this process, pyruvate undergoes transamination with ammonia, produced from AA catabolism, to form Ala. Alanine is then transported to the liver, where its nitrogen enters the urea cycle, while the pyruvate is converted back into glucose (Felig, 1973; Frayn, 2010). Additionally, the release of Gln from AT further supports its role in interorgan nitrogen transfer (Owen et al., 2002; Frayn, 2010), potentially supporting lactogenic processes in the mammary gland. However, skeletal muscle remains the primary tissue for releasing Gln and Ala during energy restriction, and the quantitative role of AT in this process is unclear in dairy cows. Consistent with these findings in AT, muscle metabolomics in the same animals demonstrated pronounced proteolysis at calving, reflected in an overall rise in free AA concentrations, except for Gln (Sadri et al., 2025). Thus, the increased demands of early lactation are met not only by enhanced release from skeletal muscle, the primary site of synthesis and storage (Newsholme and Parry-Billings, 1990; Meijer et al., 1995), but also likely through mobilization from AT.

The decrease in Gln is particularly noteworthy given its multifaceted roles during the periparturient period. It is known from obese humans that AT inflammation is linked to reduced glutamine synthetase expression and decreased Gln concentrations (Petrus et al., 2020). Even though the cows in the current study were not overconditioned, systemic inflammation with the release of cytokines and acute phase proteins (**APP**) is well documented as a pathophysiological state of postpartum cows (Trevisi et al., 2025). Locally in AT, increased expression of APP was documented around calving (Saremi et al., 2012, 2013). Serving as a critical substrate for nucleotide biosynthesis and as a precursor for glutathione synthesis, its reduction in AT also likely reflects mobilization from tissue stores to fuel immune responses and bolster antioxidant defenses (Petrus et al., 2020), probably also during the stress of calving.

AA-related Metabolites and Biogenic Amines

The heatmap in Figure 3A displays the temporal changes of AA-related metabolites and biogenic amines in AT across sampling points. The overall pattern shows relatively stable concentrations during the dry period (wk −5 to wk −1), whereas divergent patterns were observed both before and after the dry period, indicating a metabolic reprogramming of AT in response to the cessation and the resumption of lactation. The volcano plot comparing wk −7 to wk −5 (Figure 3B) revealed significant increases in choline, betaine, and phenylacetylglycine (**PAG**). Choline, a key component of structural lipoproteins and membrane lipids and the precursor for acetylcholine, is oxidized to betaine (Ueland, 2011). This oxidation links choline and betaine through the betaine-homocysteine methyltransferase reaction (Selhub, 1999), integrating them into folate-dependent one-carbon metabolism (Arumugam et al., 2021). Thus, the increase in betaine and choline, as essential methyl donors and precursors for phospholipid synthesis, may support enhanced lipid mobilization and membrane reorganization during this preparatory phase. Notably, in the companion studies, similar rises in betaine were observed in skeletal muscle over the dry period (Sadri et al., 2025) and in liver (Ghaffari et al., 2024b), indicating a coordinated mobilization of methyl donors across tissues. Phenylacetylglycine, a gut microbiota-derived metabolite, is primarily formed when gut microbes convert dietary Phe into phenylacetic acid, which is then conjugated with Gly by host enzymes in the liver. In rats, this metabolite has been linked to alterations in lipid metabolism (Delaney et al., 2004), and recent studies in humans suggest it may offer mechanistic insights into how gut Phe metabolism could influence cardiovascular risk (Zhu et al., 2023). Although direct evidence for the role of PAG in AT remains limited, the increase in PAG observed in the current study may indicate enhanced aromatic AA turnover or shifts in the gut microbiome-AT axis that prime AT for upcoming metabolic challenges. It is also likely that some of the observed changes in gut microbiota-derived metabolites are attributable to dietary modifications made during the dry-off period (Daniel et al., 2022).

In contrast, the transition from wk −1 to wk 1 was associated with declines in taurine, PAG, betaine, carnosine, and p-cresol sulfate (**p-cresol-SO$_4$**). These decreases likely indicate increased utilization of these metabolites during early lactation. The results further support a phase-specific metabolic adaptation, where AT accumulates specific methyl donors and AA derivatives in late dry-off to likely facilitate lipid mobilization and membrane reorganization during this preparatory phase. These metabolites are then rapidly depleted immediately postpartum. For example, lower taurine levels may indicate increased consumption to bolster antioxidant defenses amid the heightened oxidative stress associated with lactation. Indeed, taurine's antioxidant role, particularly in maintaining mitochondrial function, has been well documented (Jong et al., 2021). The decline in taurine levels in AT contrasts with its increase in the liver, as observed in the companion study (Ghaffari et al., 2024b), underscoring tissue-specific functions, such as bile acid conjugation in the liver versus its distinct role in AT. Furthermore, the declines in PAG and p-cresol-SO$_4$ may reflect altered aromatic AA metabolism and modulation of gut microbial interactions during this critical metabolic shift. P-cresol sulfate is produced when bacteria from the *Coriobacteriaceae* or *Clostridium* genera ferment Tyr (Saito et al., 2018), followed by sulfation by the host's cells (Wikoff et al., 2009). This sulfation process is part of phase II detoxification, facilitating the renal elimination of harmful toxins (Viaene et al., 2013). Despite known microbiota-related metabolite changes, such as PAG and p-cresol-SO$_4$, their effect on AT metabolism in dairy cows remains unclear. Notably, administration of p-cresol-SO$_4$ to healthy mice has been linked to insulin resistance and lipid redistribution in muscle and liver tissues (Koppe et al., 2013). It is unclear whether the observed decrease in gut microbiota-associated metabolites in AT immediately postpartum is due to gut microbiome shifts induced by dietary changes or alterations in renal function. Nonetheless, these findings underscore the need to investigate gut-AT interactions in periparturient cattle. These declines mirror changes seen in other tissues of the companion study (Ghaffari et al., 2024b), where liver also showed significant reductions in carnosine and p-cresol sulfate after parturition. These results further support a phase-specific metabolic adaptation across tissues and suggest potential gut-AT crosstalk in periparturient dairy cows.

Acylcarnitines

Despite significant metabolic reprogramming observed across sampling time points, the AT acylcarnitine heatmap (Figure 4A) shows largely stable patterns when comparing wk −7 and −5. Close to calving (wk −1 to 1), changes were observed but did not reach significant levels. Adipose tissue primarily functions as a storage site for FA rather than a major site of oxidation. Acylcarnitines were detectable in AT, and their presence should be linked to

FA metabolism, as acylcarnitines play a crucial role in transporting FA into mitochondria for β-oxidation and energy production (Flanagan et al., 2010; Schooneman et al., 2013). In bovine adipocytes, mitochondria may have the capacity for FA oxidation, similar to human white AT (Heinonen et al., 2020; Yang and Mottillo, 2020). However, the esterification of FA with carnitine in AT may also serve additional functions. This process could act as a temporary repository for substrates intended for FA synthesis, function as a buffering mechanism to prevent toxic fatty acyl-CoA accumulation, or help maintain a balanced cellular CoA pool (Virmani and Cirulli, 2022). Acylcarnitine metabolism undergoes considerable alteration primarily in metabolically active organs such as the liver and muscle, which are involved in intensive FA oxidation during lactation (Grummer, 1995; Kuhla et al., 2011; Schäff et al., 2013). This is supported by dynamic changes in acylcarnitine profiles observed in the liver (Ghaffari et al., 2024b) and muscle (Sadri et al., 2025) in companion studies, as well as in our previous research on periparturient dairy cow muscle (Yang et al., 2019).

Acylglycerols

Figure 5A presents a heatmap of AT TG and diacylglycerol (**DG**) species across sampling time points, revealing that the majority of TG and DG profiles remained relatively stable over time. Interestingly, volcano plots comparing different time points show that DG (16:1_16:1) concentrations in AT exhibited a biphasic pattern across the transition period. Specifically, DG (16:1_16:1) decreased progressively from wk −7 to −5 and from −5 to −1 but increased significantly at wk 1 compared with wk −1. Moreover, DG (18:1_18:1) also showed a significant decrease when comparing wk −1 to −5. In the course of energy metabolism, DG play a dual role: as precursors of TG synthesis by DG acyltransferase and as a product of TG hydrolysis mediated by adipose triglyceride lipase (Zechner et al., 2012). Moreover, DG serve as a basic structure of phospholipids (Sakane et al., 2018) and function as signaling molecules that activate proteins containing DG-responsive elements, such as protein kinase C (Nakano and Goto, 2022). Bovine backfat AT exhibits a higher percentage of unsaturated FA (16:1 and 18:1) than internal depots (perirenal and omental), which have a higher percentage of saturated FA (14:0, 16:0, 18:0; Pothoven et al., 1974). Thus, the coordinated decrease of both DG (16:1_16:1) and DG (18:1_18:1) may underscore their preferential channeling toward membrane remodeling or re-esterification into TG and thus the adaptation of AT to the metabolic challenges of lactation. Furthermore, the postpartum rise in DG (16:1_16:1) likely indicates enhanced lipolysis, as reflected by a noticeable increase in circulating FA in this study (Daniel et al., 2022), with TG breakdown generating DG as intermediates (Zechner et al., 2012). The AT acylglycerol response contrasts markedly with the serum metabolome profiles reported for the same transition period in the companion study (Ghaffari et al., 2024a). Unlike the expected increase, total serum TG concentrations remained low after parturition, likely due to enhanced clearance of liberated FA by peripheral tissues, particularly the mammary gland, and the limited hepatic VLDL secretion postcalving (Ghaffari et al., 2024a).

PC

Our study demonstrates that PC profile in the AT is dynamically remodeled as dairy cows transition from late gestation into early lactation (Figures 6A–6C). Between wk −5 and −1, several diacyl-PC (**PC aa**) species, including PC aa C38:1, C40:3, C32:1, C30:0, C40:4, and C32:0, rose significantly (Figure 6B). In parallel, specific alkyl-acyl-PC (**PC ae**) species (PC ae C34:3 and C34:2) and the lysophosphatidylcholine (**LysoPC**) a C18 were also increased. Notably, however, PC aa C38:3 and C38:4 along with PC ae C38:3 decreased during this period (Figure 6B). Phosphatidylcholine is the predominant phospholipid in mammalian AT; its stereospecific FA distribution is key for maintaining membrane fluidity and function (Body, 1988). The early adaptive increase in specific PC species during the late dry period may serve to stabilize adipocyte membranes and also the phospholipid monolayer surrounding the lipid droplets within the adipocyte (Wölk and Fedorova, 2024) and prepare them for the intense lipolytic and lipid transport demands of lactation. At the onset of lactation (comparing wk 1 to −1), a broad decline in PC species became evident. This decline was particularly evident in numerous diacyl-PC species, including C30:0, C30:2, C32:1, C36:3, C36:4, C36:5, C38:3, C38:4, C38:5, C38:6, C40:4, and C40:5, which declined markedly (Figure 6C). Similarly, a significant decrease was observed in several alkyl-acyl-PC species, such as PC ae C30:1, C30:2, C34:0, C34:1, C36:1, C36:2, C38:1, C38:3, C40:3, C40:5, C40:6, and C42:2, in addition to a reduction in LysoPC a C26:0 (Figure 6C). Thus, the marked decline in PC at the onset of lactation may suggest their repurposing in response to the heightened metabolic stress of early lactation, potentially for energy production, modulating cellular signaling through changes in lipid raft organization, or facilitating elevated phospholipase-mediated catabolism. Supporting this notion, studies in rodent and human models have linked shifts in PC composition to changes in insulin receptor signaling and overall metabolic adaptation (He et al., 2023). Future investigations targeting key enzymatic regulators such as LysoPC acyltransferase 3 (He et al., 2023) may further clarify the mechanisms underlying these adaptive phospholipid changes in bovine AT. In our companion study (Ghaffari et al., 2024b), we observed that several diacyl-PC species increased in the liver during the dry period, followed by a decline after calving. This consistent pattern across tissues supports the notion that both the liver and AT actively adjust their membrane composition to navigate the metabolic demands of drying off and calving.

SM

Our results indicate that despite substantial metabolic adaptations from late gestation to early lactation, SM levels in AT remain remarkably stable (Figure 7). Concentrations were largely unchanged from wk −7 to −5, with only minor fluctuations during the dry period (wk −5 to −1) and no significant shift at calving (wk −1 to 1). This preservation of SM levels in dairy cow AT during the transition from lactation cessation to lactation resumption may help maintain membrane integrity and lipid raft organization, crucial for proper cell signaling and lipid mobilization during rapid fluctuation in energy demands. Stringent control of SM may also prevent excessive ceramide accumulation (generated via the enzymatic hydrolysis of SM by sphingomyelinases; Kolak et al., 2012), which influences signaling pathways related to apoptosis, inflammation, and insulin sensitivity in mice (Samad et al., 2006), humans (Blachnio-Zabielska et al., 2012; Kolak et al., 2012), and cattle (Leung et al., 2020; Kenéz et al., 2022; Zhao et al., 2024). These data may suggest that not all lipid classes are equally affected by lactational stress; rather, some lipid pools may be buffered to sustain cellular homeostasis. Given that SM contribute to lipid raft formation and proper membrane receptor function, their stability may be essential for maintaining efficient cellular communication during dramatic shifts in energy demand.

The relatively limited alterations in SM observed in this study may be somewhat unexpected, given prior evidence linking these lipid classes to insulin resistance and other key metabolic pathways (Rico et al., 2015, 2018). It is important to consider that our analyses focused on subcutaneous AT, which represents only one component of the overall AT network. Visceral AT is recognized as being more metabolically active than subcutaneous depots (Ji et al., 2014; Kenéz et al., 2015; Contreras et al., 2017) and may therefore undergo more pronounced lipid remodeling during the transition period in dairy cows. However, longitudinal sampling of visceral AT in dairy cows is technically challenging, which limits the feasibility of such studies. It is therefore possible that the more substantial alterations in these lipid classes may occur primarily within visceral depots, and this should be considered when interpreting the current findings.

CONCLUSIONS

Our study represents the first comprehensive report characterizing the AT metabolome in Holstein dairy cows during the critical transition from the cessation to the resumption of lactation. During the early dry period, from wk −7 to wk −5, the AT metabolome exhibited only minor changes, suggesting that the metabolic state of the AT is relatively stable during this phase. However, significant metabolic reprogramming became evident closer to calving, particularly from wk −1 to wk 1. Distinct metabolomic profiles revealed shifts in AA metabolism, with decreased Ala, Asp, and Gln between wk −1 and 1, likely due to increased utilization within AT, redirecting their carbon skeletons toward glyceroneogenesis and FA re-esterification. This shift may also support interorgan nutrient exchange, with potential Ala export through the glucose-Ala cycle contributing to gluconeogenic substrate supply in the liver postpartum. Concurrent changes in methyl donors and phospholipid species indicate that the lipid membrane is actively remodeled, likely to optimize lipolysis and lipid mobilization in response to the metabolic challenges of calving. The relatively stable acylcarnitine levels confirm that AT functions as a long-term lipid reservoir rather than a site of FA oxidation, which predominantly takes place in other tissues such as liver and muscle. Together, these coordinated metabolic adaptations not only meet the energetic and biosynthetic demands of early lactation but also provide a framework for understanding the critical regulatory shifts in AT function.

NOTES

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Nonstandard abbreviations used: APP = acute phase proteins; AT = adipose tissue; BCAA = branched-chain AA; DG = diacylglycerol; FA = fatty acids; FIA-MS/MS = flow injection analysis-tandem MS; G3P = 3-phosphoglycerol; IQR = interquartile range; LC-MS/MS = liquid chromatography-tandem MS; LOD = limits of detection;

LysoPC = lysophosphatidylcholine; p-cresol-SO_4 = p-cresol sulfate; PAG = phenylacetylglycine; PC = phosphatidylcholines; PC aa = diacyl-PC; PC ae = alkyl-acyl-PC; PCA = principal component analysis; PERMANOVA = permutational multivariate ANOVA; PLS-DA = partial least squares discriminant analysis; SM = sphingomyelines; TCA = tricarboxylic acid; TG = triacylglycerol.

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